

Solution Architecture

Date	05 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture for the Personalized Networking Assistant provides an AI-powered framework for helping users build meaningful professional connections. Its goals are to:

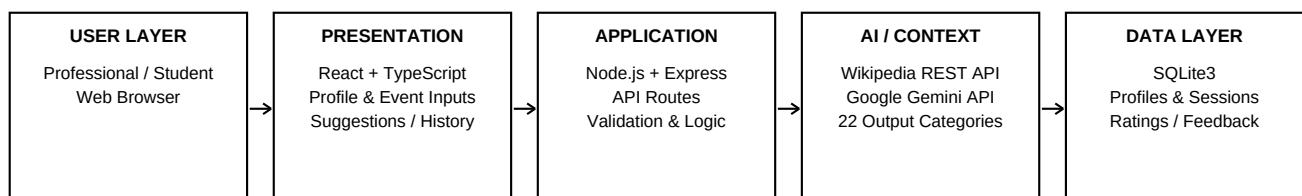
Solution Architecture:

The Personalized Networking Assistant uses a layered web architecture to deliver secure, context-aware networking assistance. Its goals are to:

1. Provide a responsive web interface for profiles, event context, AI suggestions, history, feedback, and analytics.
2. Process requests through a Node.js and Express backend with application logic and persistent SQLite storage.
3. Use the Wikipedia REST API for factual context and Google Gemini for personalized networking content.

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Example - Solution Architecture Diagram:



Request Flow: User → React UI → Express API → Wikipedia context → Gemini generation → SQLite persistence → UI response

DEPLOYMENT: Local Development → GitHub Repository → Render Production Deployment → Public Web Application

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the responsive user interface for registration, profile setup, event and topic input, generated networking content, history, feedback, and dashboard views.	React, TypeScript, Tailwind CSS
Application / API Layer	Handles API requests, validation, business logic, profile and event processing, AI orchestration, session handling, and response delivery.	Node.js, Express
AI & Context Services	Retrieves factual event context from Wikipedia and generates personalized conversation starters and related networking content.	Google Gemini API, Wikipedia REST API
Data Layer	Stores user profiles, application sessions, generation history, ratings, likes/dislikes, feedback, and dashboard metrics.	SQLite3, JSON
Deployment Layer	Hosts the production application and connects the deployed service to the GitHub-based project workflow.	Render, GitHub